

Mihal Dimo

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EDUCATION

University of Catania

2021 – 2026

Bsc in Computer Science

Relevant Modules: **Embedded systems, Algorithm & Data Structures, Linear Algebra**

Brandenburgische Technische Universität Cottbus

2024 – 2024

Bsc in Computer Science

Exchange Study Erasmus+

Relevant Modules: **Applied Linear Algebra for AI, Calculus, Software Security**

WORK EXPERIENCE

Software Engineer (Open Source)

07/2026 – Present

vLLM Project

Open Source Contributor

- Engineered **DSpark/EAGLE-style speculative decoding** for the vLLM Apple Silicon Metal backend, with ongoing optimizations to draft generation and token acceptance.
- Optimized **speculative decoding** by introducing scheduler-managed KV prefix reuse and eliminating redundant KV re-ingestion, reducing **8K first-propose latency** $31\times$ (**3.47s** $\rightarrow$ **112ms**), E2E latency **2.1** $\times$ (**64.0** $\rightarrow$ **30.8 ms/token**), and steady-state KV ingest **50%** (**4** $\rightarrow$ **2 tokens**).
- Extended paged **KV-cache** infrastructure with configurable memory usage reporting, improving runtime diagnostics and memory planning for long-context inference workloads.

Software Development Engineer Intern

01/2026 – 07/2026

Amazon

Berlin, Germany

- Engineered a **full-stack embedding search platform** over **50M embeddings**, with PCA/UMAP visualization with under **5 ms K-Nearest Neighbours** queries, enabling faster Music recommender experimentation.
- Designed a **two-stage batch indexing pipeline**: a weekly job performs metadata hydration and writes partitioned **Parquet files to S3**, followed by an **AWS ECS ingestion task** that sequentially indexes entities into OpenSearch.
- Implemented a **Parquet-based storage layer on S3** with entity-level partitioning, reducing per-session data transfer by $\sim 20\times$ and improving ML experimentation efficiency.
- Developed a **FastAPI backend on AWS ECS** that unifies search, ingestion status, and visualization workflows, with **CloudWatch monitoring** for pipeline reliability.

Firmware Lead & Embedded Software Engineer

03/2024 – 12/2025

KBDFans (Partner)

Changzhou, Jiangsu, China

- Lead firmware and hardware development for the **Zellia Hall Effect project** (team of 5 C/C++ engineers), designing a **distributed embedded system** across 5 AT32 MCUs with enhanced modularity and multi-layout keyboard using **KiCad 8.0**.
- Achieved **<0.28 ms input latency** with **106 kHz Scan Rate** by decoupling signal acquisition and processing, and engineered a custom **7.5 Mbps UART protocol** for high-throughput multi-MCU synchronization.
- Offloaded **ADC normalization** and calibration to slave MCUs, reducing master CPU load by **90%** and enabling faster processing.
- Developed a cross-platform configuration tool using **SvelteKit**, and **TypeScript**, enabling device tuning, firmware updates and profile management with 8+ configurable profiles.

PROJECTS

Deepseek Code - Terminal AI Coding Agent

- Engineered a terminal **AI coding agent** in **TypeScript** with a multi-step agentic loop, real-time streaming, tool execution, and **MCP** extensibility, powered by the in-house **AI SDK C++**.
- Implemented a provider abstraction layer supporting **OpenAI-compatible APIs** and integrated **MCP** for runtime model switching, server discovery, and third-party tool execution.

AI SDK C++ — High-Performance Agent Framework

- Developed a **C++20 LLM orchestration framework** using coroutines and Boost.Asio for high-concurrency, non-blocking AI agent workflows across multiple providers.
- Implemented a **C ABI FFI** with Python, Node.js, Rust, and Go, enabling reusable tool execution, context management, and rate-limit handling for autonomous multi-step agent systems.

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, TypeScript, Java, SQL.

ML Systems: vLLM, MLX (custom Metal kernels), PyTorch, Metal/MPS backends, speculative decoding, quantization, KV-cache optimization, embeddings, recommender systems, DQN.

LLM Tooling: MCP, multi-provider LLM APIs, AI-SDK, Claude code.

Systems & Data: Linux, Docker, AWS, REST APIs, Kafka, PySpark, OpenSearch, Parquet, Pandas, Numpy.

Embedded & Web: ARM Cortex-M, KiCad, USB/WebUSB, FastAPI, Node.js, SvelteKit, React.